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9648 / 02

2 hours

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

This document consists of 2 sections:

Answer **ALL** questions.
Answers are to be written in the spaces provided.

Answer **ONE** question.
Write your answers on the separate writing papers
provided.

For Examiner's Use	
Structured Questions	
1	/ 9
2	/ 10
3	/ 11
4	/ 10
5	/ 10
6	/ 10
7	/ 10
8	/ 10
9 or 10	/ 20
Total	/ 100

Do not open this booklet until you are told to do so.

Section A: Structured Questions [80 marks]

Answer all the questions in the space provided.

Question 1 [9 marks]

Enzyme X digests a sugar. Figure 1.1 shows the rate of reaction of enzyme X and a mixture of enzyme X and substance Y on the sugar. Both reactions were carried out in a conical flask.

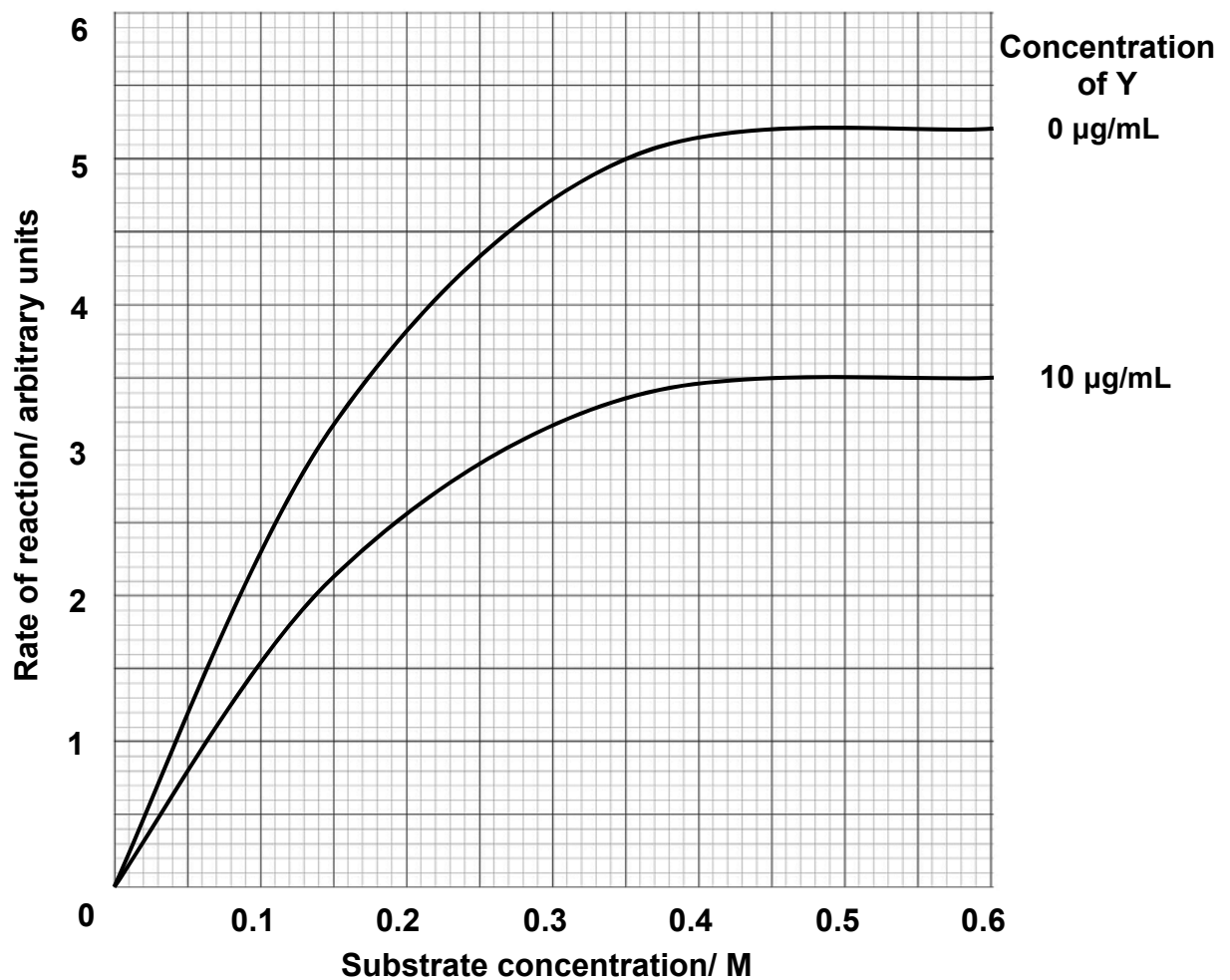


Figure 1.1

- (a) (i) Describe and explain the effect of substance Y on the rate of reaction catalyzed by enzyme X. [3]

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- (ii) On Figure 1.1, draw the rate of reaction catalyzed by enzyme X in the presence of **5 $\mu\text{g/mL}$ of Y**. [1]

Enzyme X may be immobilized in alginate beads. Figure 1.2 shows a comparison between the activities of free enzyme X and immobilized enzyme X over a range of temperatures. Equal concentration of free enzyme X and immobilized enzyme X were used.

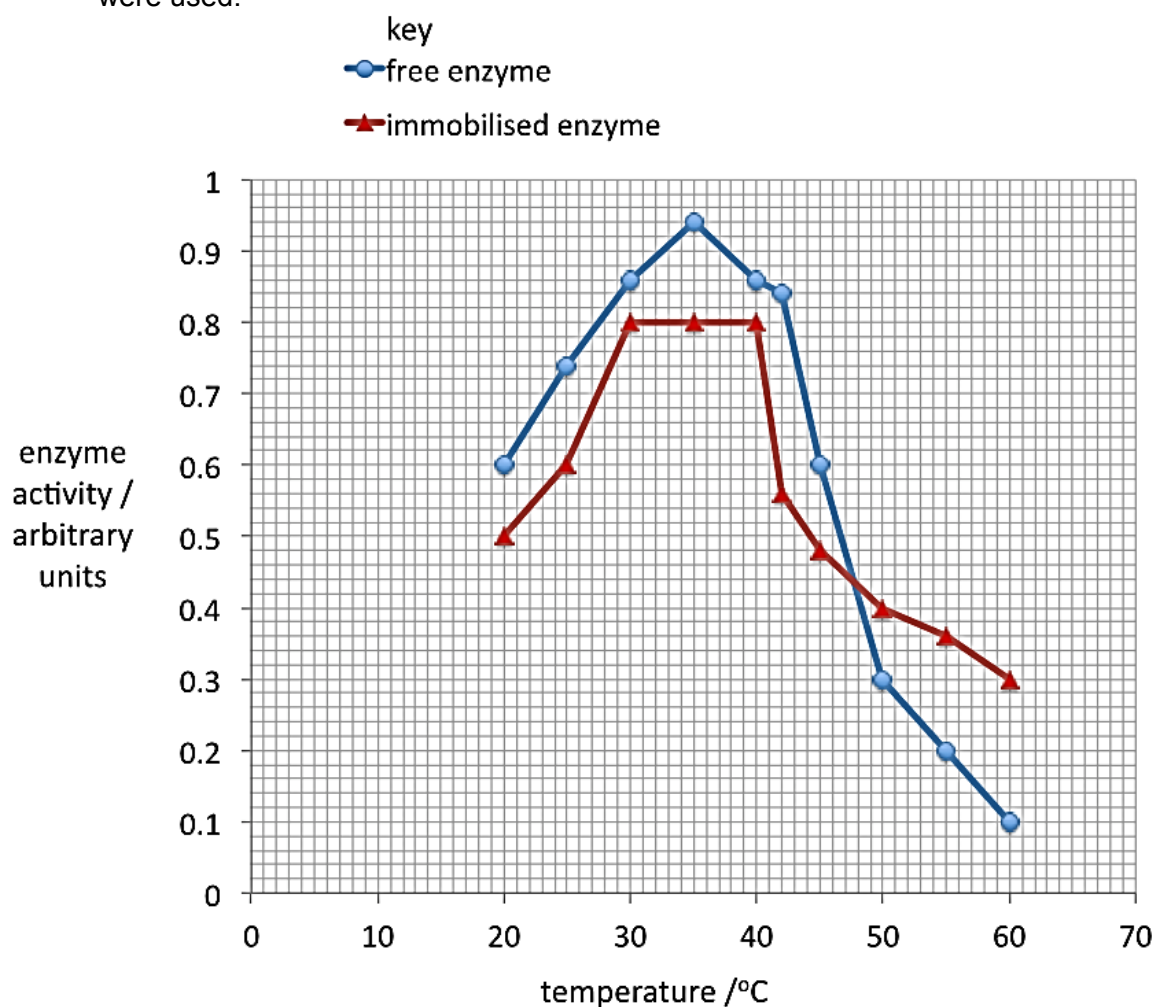


Figure 1.2

(b) With reference to Figure 1.2,

(i) describe and explain the effect of temperature on the activity of free enzyme X.
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(ii) describe and explain the differences between the activity of free enzyme X and immobilized enzyme X up to 40°C. [2]

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Question 2 [10 marks]

The control of gene expression functions in a wide range of developmental processes including sex determination.

In *Drosophila melanogaster*, the doublesex (dsx) protein is crucial in the development of sexual phenotype. In males, the presence of dsx-m increases the transcription of genes whose products control male sexual development. The transcription of these genes are repressed in the presence of an isomer dsx-f which is found in females.

Figure 2.1 shows the structure of the dsx-m protein which has a DNA-binding domain and protein-binding domain.

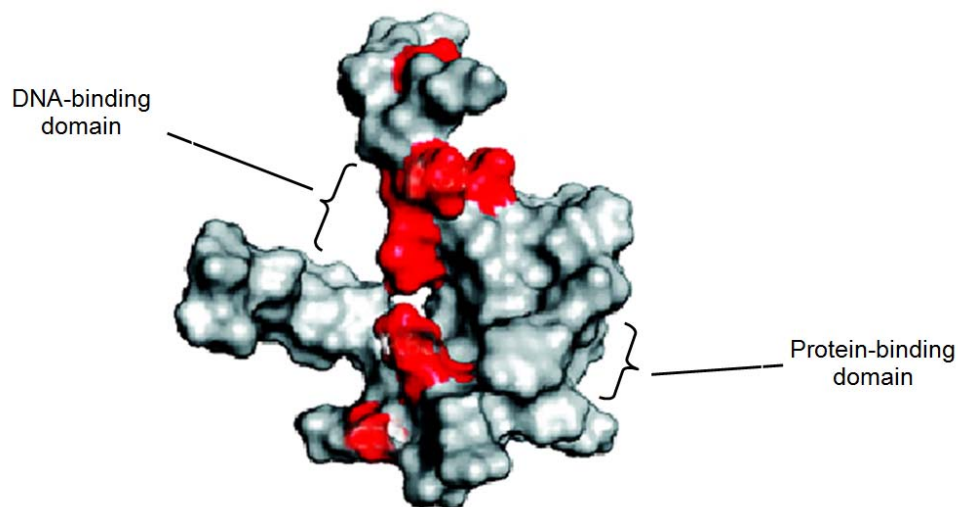


Figure 2.1

(a) State the role of proteins dsx-m and dsx-f. [1]

dsx-m:

dsx-f:

It was found that male *Drosophila* flies showing female characteristics mostly have an amino acid substitution at the DNA-binding domain of the dsx-m protein.

The shaded areas in Figure 2.1 show the positions of amino acid substitutions presumed to affect the DNA-binding domain.

- (b) Explain why dsx-m protein contains both a DNA-binding domain and a protein-binding domain. [2]

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- (c) Explain why an amino acid substitution within the DNA-binding domain can affect the function of dsx-m protein. [2]

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Proteins dsx-f and dsx-m are involved in the regulation of *Drosophila* *yolk protein* gene. The *Drosophila* *yolk protein* gene is expressed in the fat body of adult female under the control of *dsx*. The presence of dsx-f activates the transcription of *Drosophila* *yolk protein* gene, which allows *Drosophila* *yolk protein* gene to be expressed in the fat body of adult female. The presence of its isomer, dsx-m in males represses the transcription of *Drosophila* *yolk protein* genes.

An *in vitro* transcription under the control of a eukaryotic promoter of *Drosophila* *yolk protein* gene has been developed. The transcription of DNA can occur in:

- **Nuclear extract:** with contents of cell nucleus without genome from a female *Drosophila*
- **Purified system:** with purified RNA polymerase and the following general transcription factors, TFIID (the TATA binding factor), TFIIB and TFIIIE (which bind to RNA polymerase).

A series of experiments that compare the efficiency of transcription in the purified system with the efficiency of transcription in a crude nuclear extract have been performed. The two systems were tested with the undeleted template DNA and with various deleted templates that were generated as shown in Figure 2.2. The number represents the number of bases away from the transcription start site (+1).

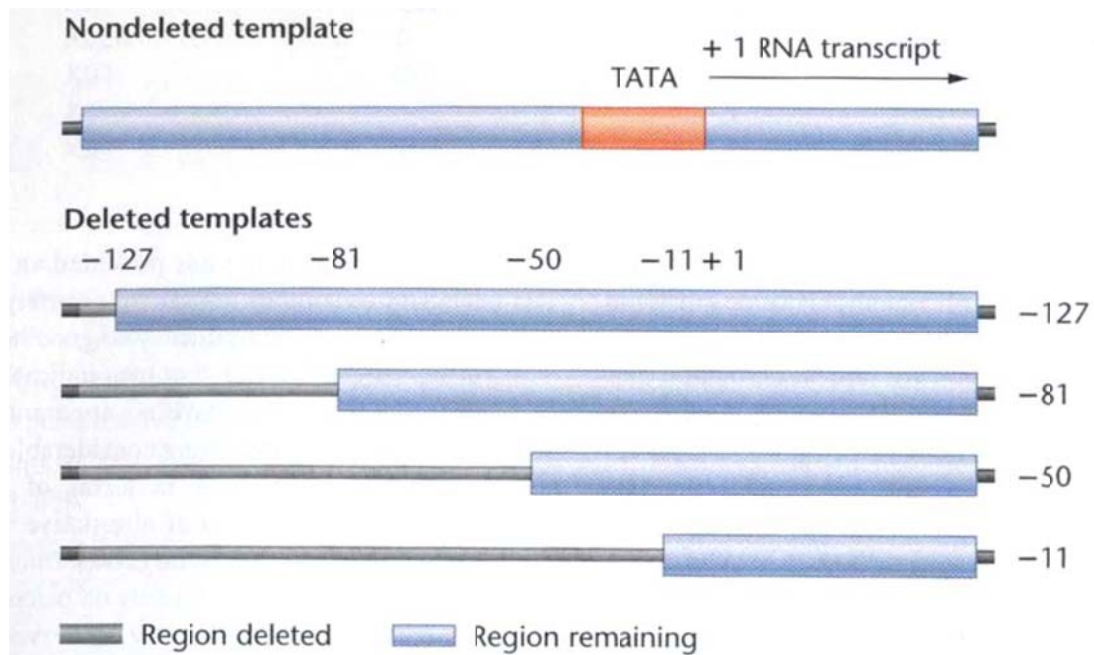


Figure 2.2

Results of the test are shown in Table 2.3.

DNA template added	Nuclear extract	Purified system
undeleted	++++	+
-127 deletion	++++	+
-81 deletion	++++	+
-50 deletion	+	+
-11 deletion	0	0

Table 2.3

Legend:

- + Low efficiency transcription
- ++++ High efficiency transcription
- 0 No transcription

(d) Explain why there is no transcription from the -11 deletion template. [1]

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(e) With reference to Table 2.3,

- (i) explain the differences in the results between the nuclear extract and the purified system for the undeleted template. [2]

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- (ii) explain the difference in the results between the -50 deleted template and -81 deleted template in the nuclear extract. [2]

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Question 3 [11 marks]

Galactose (*gal*) operon is shown in Figure 3.1. The *gal* operon is regulated negatively by a repressor in a manner comparable to *lac* regulation. The *galactose* operon of *E. coli* consists 3 structural genes - *galE* (epimerase), *galT* (galactose transferase) and *galK* (galactokinase), which are transcribed from two overlapping promoters P_1 and P_2 upstream from *galE*. Regulation of the operon is complex since the *galE* product, an epimerase that converts UDP-glucose into UDP-galactose, is required for the formation of UDP-galactose for cell wall biosynthesis.

When galactose is present, there is a higher rate of transcription to produce enzymes to metabolize galactose. The *gal* operon is controlled by cAMP-CAP as for the *lac* operon. cAMP-CAP binds to region promoter P_1 promoting transcription from P_1 but inhibiting transcription from P_2 . When cells are grown in glucose, basal level transcription occurs from P_2 .

GalR codes for a repressor protein, promoter and operators (O_E and O_I) are present in the *gal* operon.

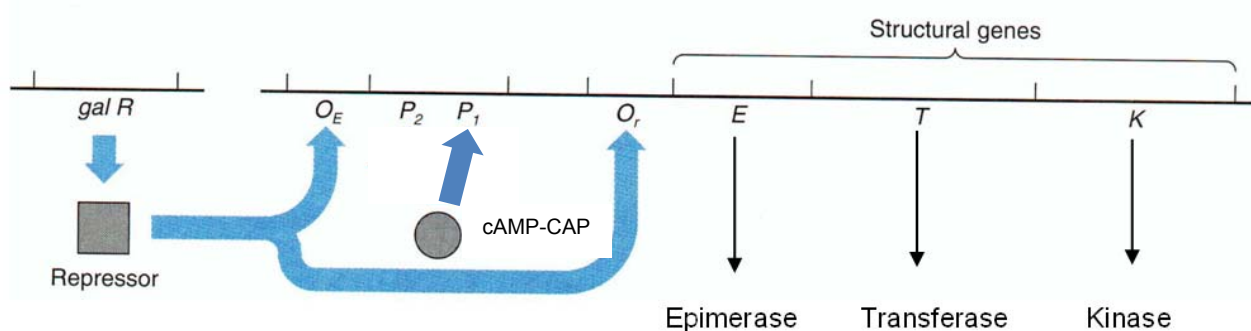


Figure 3.1

(a) In the context of *gal* operon, explain what is meant by the term *operon*. [2]

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(b) Using *gal* operon as an example, explain what is meant by *negative control*. [1]

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(c) Explain the function of galactose in this system. [2]

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(d) How would expression of the structural genes in *gal* operon be affected if gene for *galR* was missing? [1]

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(e) Describe how the *gal* operon works under high galactose and low glucose concentration conditions. [4]

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(f) With reference to Figure 3.1, suggest how binding of *gal* repressor prevents the transcription of the structural genes. [1]

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Question 4 [10 marks]

The World Health Organization estimates that 5 million children die each year as a consequence of acute diarrhea. *Escherichia coli* is the cause of a third of cases of childhood diarrhea in developing countries and is also the most prominent cause of diarrhea in travellers to developing countries. *Shigella* species, the cause of bacillary dysentery, taxonomically constitute a subspecies of *E. coli*. Dysentery is an intestinal inflammation, especially in the colon, that can lead to severe diarrhea with mucus or blood in the feces.

Effective treatment and prevention measures are lacking for *E. coli* diarrhea. The mainstay of treatment is oral rehydration. This simple and inexpensive measure has saved countless lives, but it does not influence the natural course of disease and has no intrinsic antibacterial activity. Antibiotic use is of doubtful value since resistance is widespread in *E. coli*, and vaccines are still in the early development phase. Water and sanitation programmes could improve the quality of drinking water, but are prohibitively expensive for many developing countries.

Bacteriophages have been proposed as natural antimicrobial agents to fight bacterial infections in humans, in animals or in crops of agricultural importance. In fact, phages have been used as therapeutic agents in Eastern Europe and the former Soviet Union (and before the antibiotic era in the United States). They have been administered to humans (i) orally, in tablet or liquid formulations, (ii) rectally, (iii) locally (skin, eye, ear, nasal mucosa, etc.), in rinses, and creams, (iv) as aerosols, and (v) intravenously and there have been virtually no reports of serious adverse complications associated with their use.

- (a) Identify and explain which bacteriophage would be the preferred therapeutic agent to use in the immediate treatment of dysentery for humans. [3]

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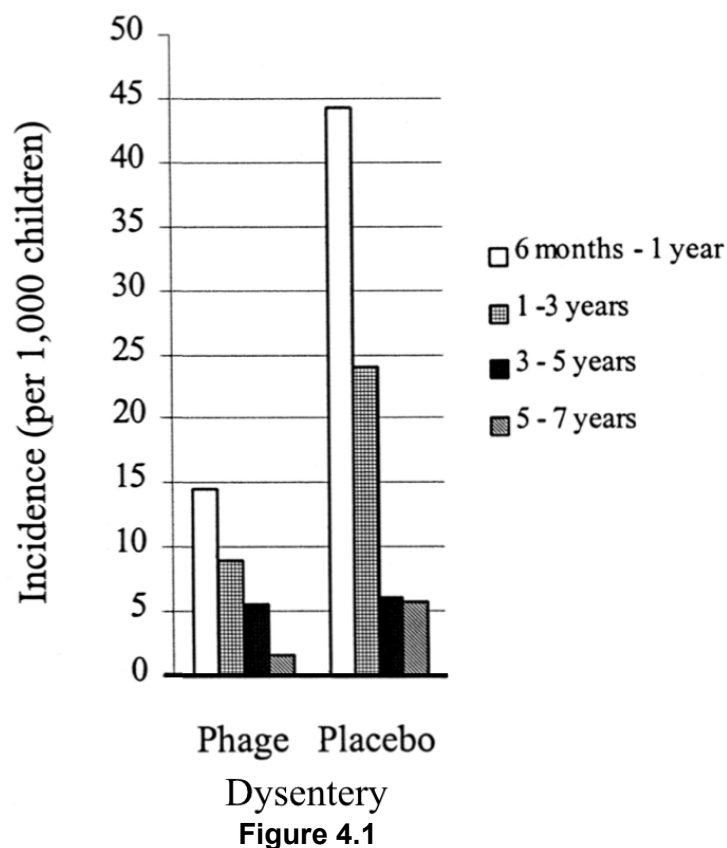
- (b) Other than the rapid development of resistant *E.coli* variants, suggest why antibiotics may not be the most effective treatment for dysentery. [1]

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One of the most extensive studies evaluating the utility of therapeutic phages for prophylaxis of infectious diseases was conducted in 1963 and involved phages against bacterial dysentery. A total of 30 769 children (6 months to 7 years old) were enrolled in an oral phage prophylaxis trial against bacterial dysentery. About half of the children living on one side of the streets received, once every week, *Shigella* phages orally in a pressed tablet form, while children on the other side of the streets received a pressed tablet made from sugar alone to act as a placebo. The children were followed for 109 days.

Figure 4.1 shows the incidence of clinical dysentery (diagnosed via appearance of initial symptoms) caused by *E. coli* other than *Shigella* sp. in phage-treated and phage-untreated (placebo) children 6 months to 7 years of age.



- (c) (i) With reference to Figure 4.1, comment on the effectiveness of phage therapy.
[3]

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(ii) Suggest the purpose of the placebo group. [1]

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Unlike bacteriophage, the influenza virus is an animal virus. A drug that can inhibit the reproductive cycle of influenza virus is oseltamivir, better known under its trademark name as Tamiflu. Its mechanism of action is illustrated in Figure 4.2 and Figure 4.3.

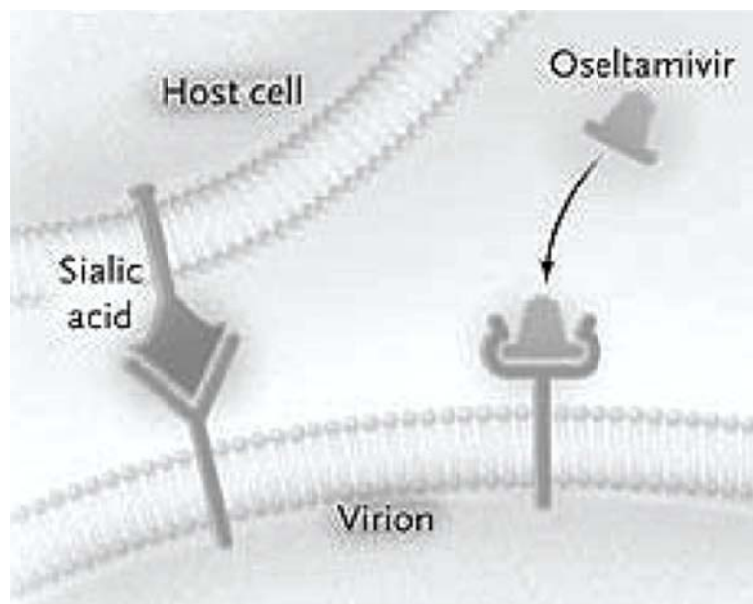


Figure 4.2

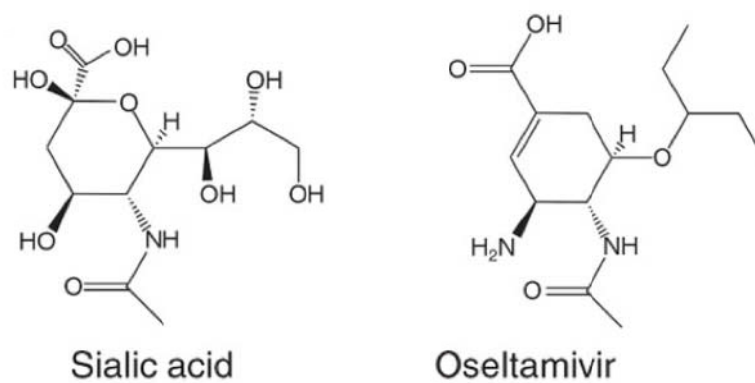


Figure 4.3

(d) Explain how oseltamivir affects the reproduction cycle of influenza virus. [2]

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Question 5 [10 marks]

The bulbs of onion plants of a particular species display a variety of skin colours. Genetically identical onion plants that produced red onion bulbs were self-pollinated. The colours and number of all the onion bulbs in the progeny were recorded.

Red bulbs 1855

Pink bulbs 634

White bulbs 782

Use the following symbols to represent the alleles:

R red r colourless

P pink p colourless

(a) (i) Draw a genetic diagram in the space below to show the cross described. [4]

(ii) Explain the term *epistasis* in this context. [3]

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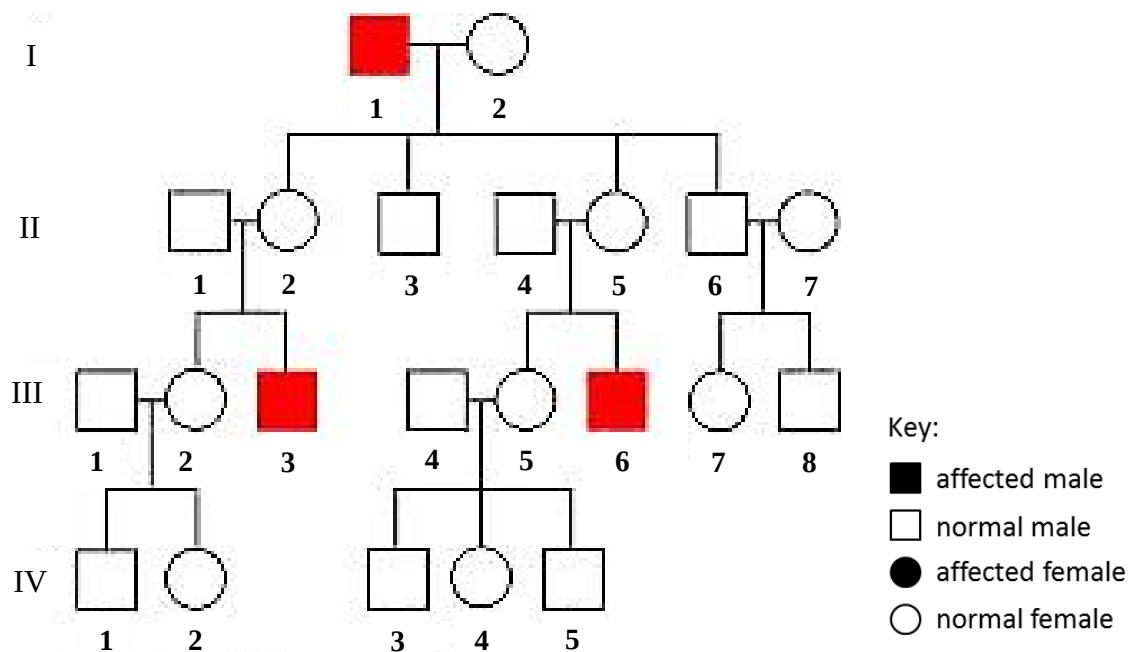
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The pedigree below shows the inheritance of glucose-6-phosphate dehydrogenase deficiency over four generations in an extended family.



(b) Explain the mode of inheritance of this disease that best explains the pedigree shown above. [3]

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Question 6 [10 marks]

In a study on the role of proton transfer through ATP synthase complex, a plant was subjected to flashes of light (indicated by the arrow) and the concentration of ATP synthesized was subsequently recorded as shown in Figure 6.1. The experiment was also conducted in the presence of venturicidin, a potent inhibitor of proton flow in ATP synthase complex.

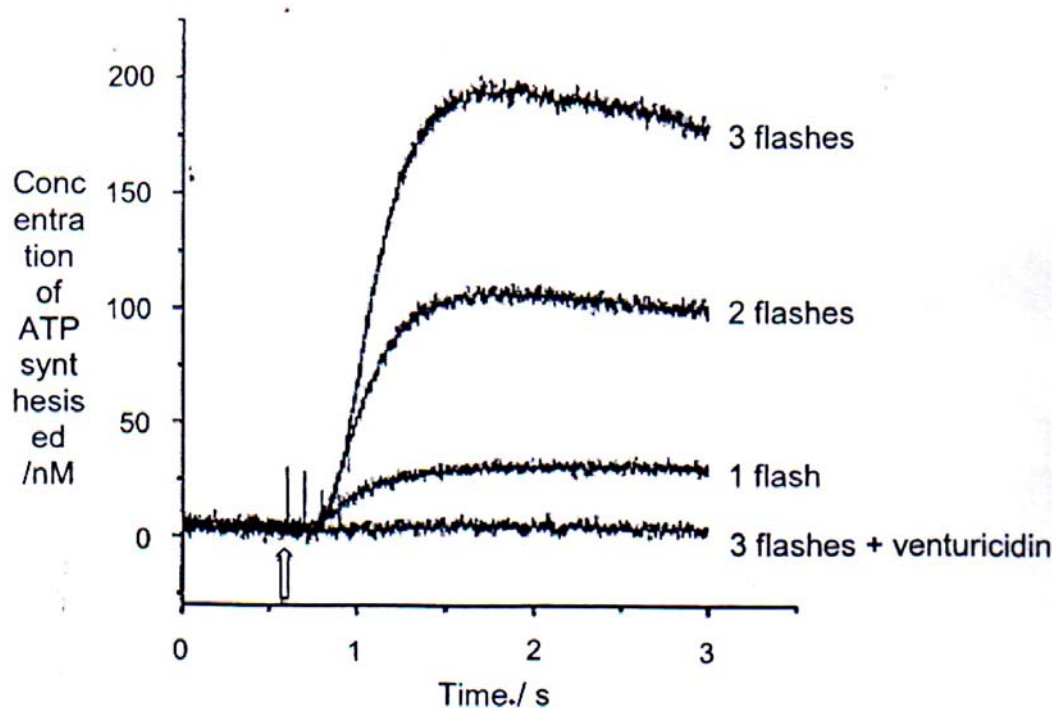


Figure 6.1

- (a) With reference to Figure 6.1,
(i) Explain the effect of flashing light on ATP synthesis in the absence of venturicidin. [3]

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(ii) Account for the effect of venturicidin on photosynthesis. [2]

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A student investigated the pathway by which carbon dioxide is converted to organic compounds during photosynthesis. He used the apparatus shown in Figure 6.2.

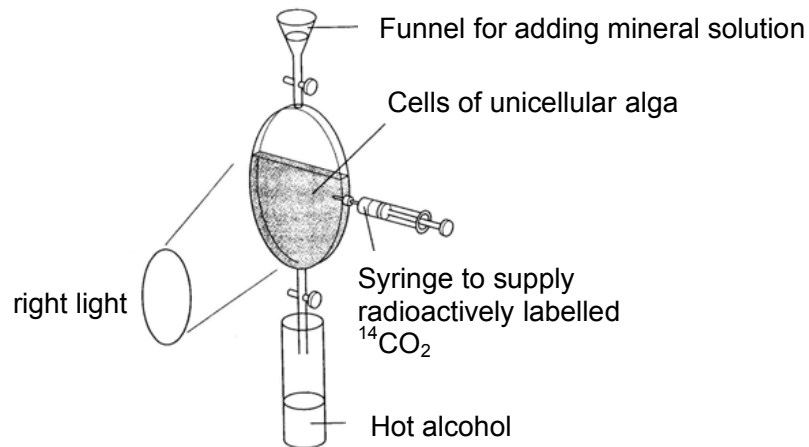


Figure 6.2

While the apparatus was in a dark room, he supplied the algal cells with $^{14}\text{CO}_2$. The contents of the apparatus were thoroughly mixed and the light was switched on subsequently. At five-second intervals, he released a few of the cells into hot alcohol, which killed the cells very quickly. The results were shown in the following chromatograph (Figure 6.3).

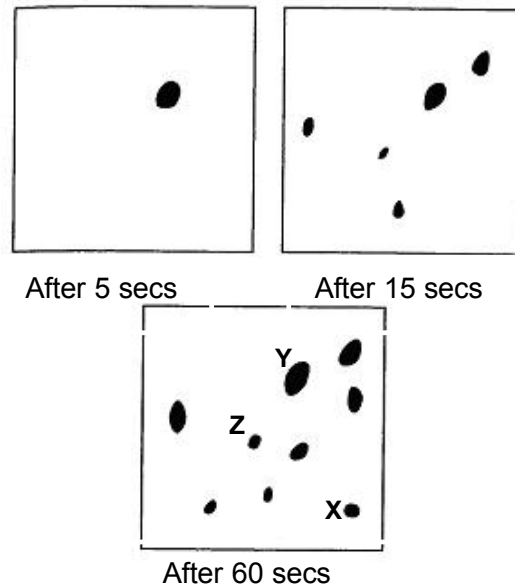


Figure 6.3

- (d) (i) One round One round of Calvin cycle takes about 60s to complete. The intermediates of the Calvin cycle were subsequently analysed to be ribulose-1,5-bisphosphate, glyceraldehyde-3-phosphate and 3-phosphoglycerate. With reference to Figure 6.3, identify X, Y and Z and explain your answer. [4]

X:.....

Y:.....

Z:.....

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- (iii) Suggest how increasing carbon dioxide concentration will affect the time of appearance of X, Y and Z on the chromatograph, [1]

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Question 7 [10 marks]

Ras is a GTP-binding protein that helps to broadcast signals from the cell surface to other parts of the cell to stimulate cell proliferation. If Ras function is inhibited, the cell proliferation responses normally induced by the activated receptors do not occur. Conversely, if a hyperactive mutant, Ras protein, is introduced into some cell lines, the effect on cell proliferation is the same as that induced by the binding of ligands to cell surface receptors. This hyperactive product of mutant Ras gene promotes the development of cancer. Figure 6.1 shows the sequence of events in a Ras cell cycle-stimulating pathway.

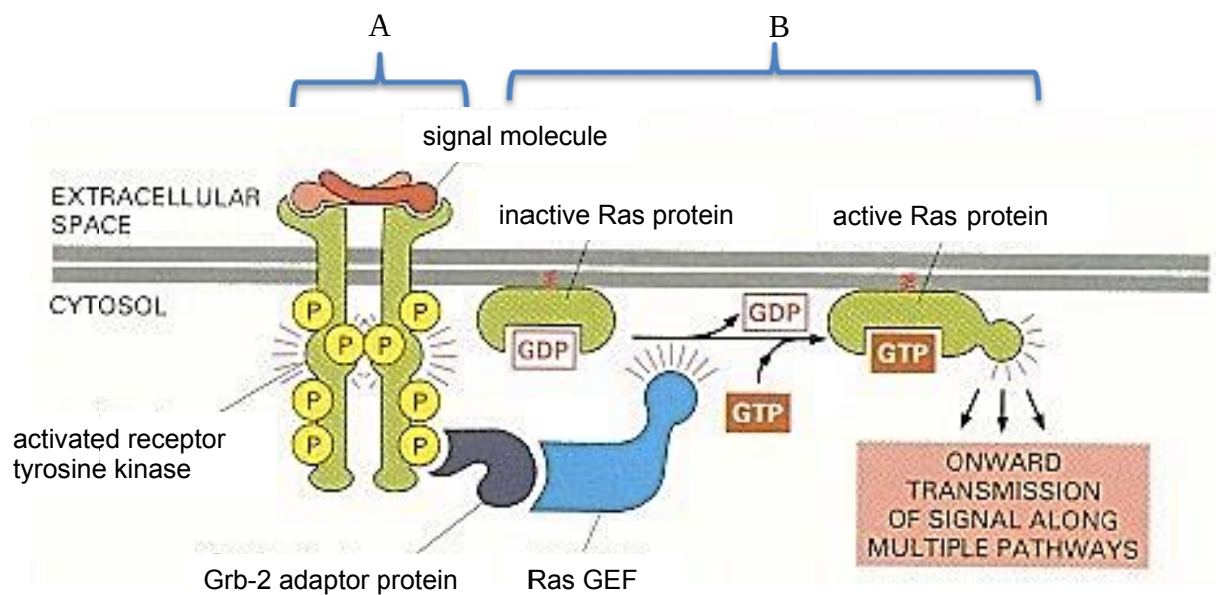


Figure 7.1

(a) (i) Briefly describe the stages A and B shown in Figure 7.1. [4]

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- (ii) Ras protein is a plasma membrane bound G-protein. Suggest how the activated G-protein can relay the signal from the ligand to activate cellular response. [2]

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The receptor for epidermal growth factor (EGF) can be activated by a deletion that removes part of its extracellular domain. These mutant receptors are able to form active dimers even in the absence of EGF, and thus they produce a stimulatory signal inappropriately. Besides mutation to the gene that codes for the extracellular domains of the receptors, 30% of human tumors have a hyperactive Ras mutation.

- (iii) With reference to Figure 7.1, suggest how mutation in the Ras gene may bring about cancer. [3]

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- (iv) Besides the abovementioned mutation, identify and suggest how a virus may cause Ras proto-oncogene to become an oncogene. [1]

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Question 8 [10 marks]

The Hawaiian islands possess one of the most remarkable insect faunas on Earth and are considered as natural laboratories for the study of the process of speciation and evolution.

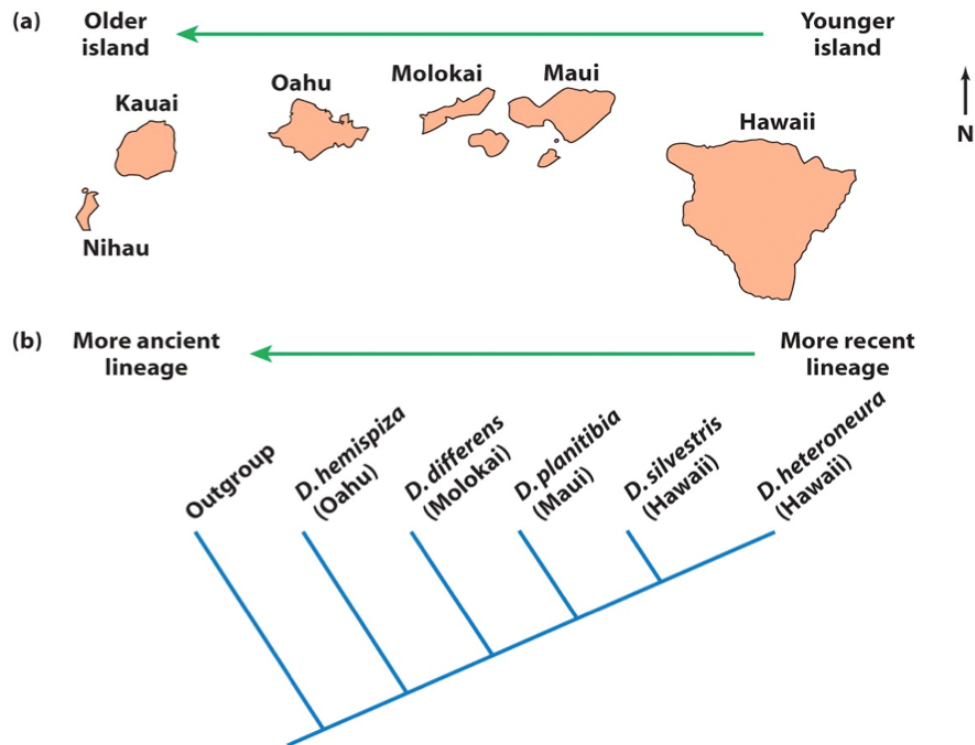


Figure 8.1 (a) Illustration of distribution and geological age of Hawaiian islands. (b). Distribution of some *Drosophila* species across the islands in correspondence how recent the species had been formed.

An example of evolution of a large number of divergent forms from a primitive ancestor in these islands can be observed from high species diversity in the genus *Drosophila* (Figure 8.1(b)). These *Drosophila* are also collectively known as Hawaiian *Drosophila*. Across the islands hold at least 411 species of *Drosophila*, with characters ranging elaborately from size, pigmented wings to different mouthparts and unique ornaments on the head (Figure 8.2). While some of these characters appear to be advantageous for survival in species-specific niches, other characters are thought to play a role in courtship behavior.

Throughout their evolutionary history, Hawaiian *Drosophila* have repeatedly colonized islands or subdivided populations within an island.



Figure 8.2

(a) Explain how the evolution of different species in the Hawaiian Islands could have occurred. [6]

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Differences in the cytochrome b DNA sequence of several species of *Drosophila* from different islands were measured and plotted against time since divergence from the primitive ancestor (MYA) as seen in Figure 8.3.

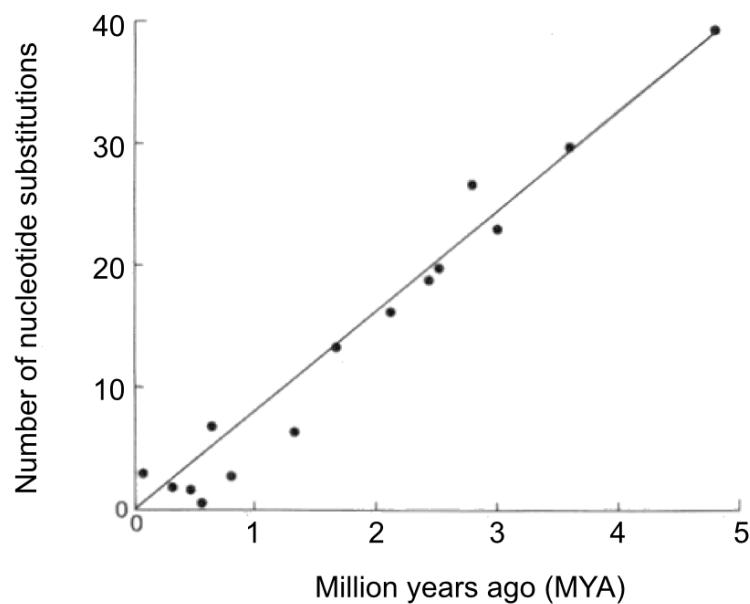


Figure 8.3

(b) (i) Describe how these changes support neutral theory of molecular evidence. [3]

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(ii) Suggest a challenge in using the molecular clock. [1]

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Section B [20 marks]

Answer **only ONE** question.

Write your answers on the separate answer paper provided. Your answers should be illustrated by large, clear labeled diagrams, where appropriate. Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 9

- (a) Explain the importance of lipids in living organisms. [7]
- (b) Compare and contrast between collagen and cellulose. [8]
- (c) Enzymes are prominent examples of globular proteins. Describe the mode of action of an enzyme. [5]

Question 10

- (a) Explain how the structure of neuron facilitates impulse transmission. [7]
- (b) Describe how unidirectional transmission of nerve impulses is ensured along a neuron and across a synapse. [8]
- (c) Describe how two named membrane proteins are involved in impulse transmission along the axon of a motor neuron. [5]